

Hong Yuan Cao

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EDUCATION

Cornell Tech, Cornell University

New York, NY

Master of Engineering in Computer Science

Expected: May 2027

Boston University

Boston, MA

Bachelor of Arts in Computer Science and Economics GPA: 3.73/4.0

Graduated: May 2026

TECHNICAL SKILLS

Languages: Python, C++, C, TypeScript, JavaScript, Java, Go, SQL, Bash, OCaml

Frameworks: React, Next.js, Node.js, FastAPI, Django, Flask, Spring Boot, PyTorch, Pandas, OpenCV, gRPC

Tools: Git, Docker, Linux/Unix, AWS, GCP, PostgreSQL, MySQL, Redis, Terraform, GDB

EXPERIENCE

Software Engineer Intern

Shanghai, China

KeelinMicro Electronics (Automotive Startup)

June 2025 – August 2025

- Authored technical draft for **Chinese patent filed Jan. 2026** on Kalman + ReID tracking (**96.3%** ID continuity)
- Built **Python/OpenCV** camera-radar fusion pipeline mapping radar detections into camera frames at **60+ FPS**
- Synchronized Pi camera and STM32 radar streams using fractional-delay **Farrow filters**, reducing skew to **<1 ms**
- Ported **V4L2/GMSL** camera capture on Raspberry Pi 5 with **C** driver changes and custom device-tree overlay

Software Engineer Intern - Databases

Shanghai, China

Tospur Real Estate Consulting Co., Ltd

June 2024 – August 2024

- Built **Django** ETL services on Alibaba Cloud to consolidate **4 MySQL databases**, cutting aggregation runtime by **45%**
- Implemented account-migration APIs and permission mapping for **200+** consultants across legacy CRM systems
- Tuned **MySQL** hot paths with indexes, batched joins, and connection pooling, cutting query latency by **23%**

Machine Learning Intern

Shanghai, China

Anban Tech (Fuzz Testing Startup)

May 2024 – June 2024

- Built **Pandas** preprocessing pipeline for JUnit corpora, normalizing test metadata and removing malformed samples
- Collected **2.2 GB** JUnit 5 dataset through the **GitHub API**; validation scripts raised valid-sample rate to **95%**

Software Engineer Intern

Shanghai, China

Sprout Technologies, Inc. – Tonic (Entertainment Tech Startup)

June 2023 – October 2023

- Shipped responsive **React** UI components and animations for Tonic marketing pages; post-launch engagement rose **24%**
- Built **Python/BeautifulSoup/Pandas** lead-generation workflow to filter **3,000+** contacts and identify **200+** targets

PROJECTS

Athens – Multi-Tenant AI Learning Platform | *FastAPI, Next.js, pgvector, AWS, Bedrock*

- Built **Cognito** onboarding, course-level RBAC, roster imports, and professor tools for **4 planned BU course pilots**
- Architected async **RAG** ingestion on ECS, Aurora pgvector, S3, and SQS/DLQ for PDFs and Blackboard exports
- Built a student workspace with persistent AI chat, private notebooks, practice exams, flashcards, and study plans
- Added tenant isolation, audit and usage controls, professor analytics, **5,500+ tests**, accessibility, and release gates

VectorDB | *C++, Cython, FastAPI, Docker*

- Built a vector database with **HNSW** ANN indexing, metadata filtering, persistence, and a **FastAPI** service layer
- Implemented native **C++/Cython** HNSW build and batch-search paths with compact **CSR** graph storage
- Benchmarked on a **100K-vector SIFT1M** subset: **98.6% Recall@10**, **4,768 queries/s**, and **0.21 ms/query**

Work-Stealing Task Runtime | *C++20, CMake, Linux*

- Built a **C++20** scheduler with Chase-Lev dequeues, generation-safe handles, and atomic DAG execution
- Reached **2.02M lightweight tasks/s** on 8 pinned cores, **3.28×** a global-FIFO pool; validated with TSan/ASan/UBSan

Loom – AI Sprite Animation Pipeline | *Go, Python, Next.js, Docker*

- Built an async **Go/Python** pipeline producing engine-ready sprites, atlases, GIF previews, and **Unity/Godot** configs
- Added provider-agnostic generation, automated quality checks, bounded retries, and traceable artifacts across the pipeline